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1-DOF Series-Elastic Finger Exoskeleton

Actuator Commissioning, Embedded Control, sEMG Intent Detection, and Virtual SEA Evaluation

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Abstract

This project investigated a one-degree-of-freedom series-elastic-actuated (SEA) finger exoskeleton for surface-electromyography (sEMG) intent detection and later sensorimotor analysis. The available Harmonic Drive PMA-8A actuator, Maxon ESCON 50/5 controller, incremental encoder, and STM32 NUCLEO-F446RE were commissioned. Closed-loop low-speed rotation was first verified using the ESCON potentiometer and then commanded by STM32 analog and digital outputs. A laptop bridge was implemented to acquire eight-channel Myo armband EMG and inertial data, calibrate a repeatable pinch-intent template, reject competing gestures and motion, log trials, and communicate with the STM32 over serial.

The physical spring, tendon transmission, finger mechanism, and spring-deflection sensor were unavailable and therefore were not represented as completed hardware. Instead, recorded real EMG classification was replayed through an explicitly labelled virtual SEA model to demonstrate the intended bounded 0–180° motion and the relationship between spring deflection and force. The result is a validated physical actuator-and-embedded-control MVP, a separately validated sEMG intent subsystem, and a reproducible simulation of the missing compliant mechanism. Physical Myo-to-actuator operation, impedance control, and electromechanical-delay measurement remain future integration work.

Contents

Chapter 1

Objective and Scope

The declared research objective was a 1-DOF dorsal finger exoskeleton driven through a series elastic element. The intended control and sensing chain was

sEMG intent → embedded controller → motor → spring/tendon → finger motion.

The proposed series spring had stiffness $k_s \approx 2.0\,\mathrm{N/mm}$. Its measured deflection would provide an indirect force estimate,

$$F_s = k_s \Delta x_s, \tag{1.1}$$

and, after mechanism calibration, an interaction torque estimate. The intended impedance law was

$$\tau_{cmd} = K_v(\theta_{target} - \theta) + B_v(\omega_{target} - \omega). \tag{1.2}$$

The final available hardware did not include the spring, Bowden/tendon transmission, displacement sensor, or printed finger mechanism. Consequently, this report distinguishes three evidence classes:

- **Measured**: voltages, ESCON monitor values, EMG, IMU, and serial responses;
- **Observed**: physical actuator motion in photographs and videos;
- **Simulated**: finger angle, spring deflection, spring force, and virtual SEA dynamics.

No simulated signal is presented as a physical measurement.

1.1 Final system architecture

The final deliverable is best understood as two validated chains joined by a conservative software interface. The low-level actuator chain was validated physically. The Myo chain was validated diagnostically with the motor disabled. The missing spring/finger mechanism was represented only in simulation.

Myo armband EMG/IMU
$\Downarrow$
Fedora laptop: calibration, template matching, motion rejection, CSV logging
$\Downarrow$
USB serial command: `0/1`
$\Downarrow$
STM32 NUCLEO-F446RE: DAC setpoint on PA4 and enable on PC0
$\Downarrow$
Maxon ESCON 50/5 velocity controller
$\Downarrow$
Harmonic Drive PMA actuator
$\Downarrow$
future SEA spring/tendon/finger mechanism

1.2 Validation ladder

The project was intentionally validated in small steps so that each risk was isolated before the next connection was made.

ESCON USB configuration
→ potentiometer-controlled actuator rotation
→ STM32 voltage-only test
→ STM32-to-ESCON button-controlled rotation
→ serial STOP/RUN/STATUS test
→ Myo diagnostic classification with motor disabled
→ virtual SEA replay from recorded detector states

Chapter 2

Laboratory Development Chronology

The final result came from a staged commissioning process rather than a single end-to-end build. This chronology is included because it shows which parts of the original SEA project were physically validated, which parts were deferred, and why the final demonstration scope was reduced.

- **17 June 2026 – hardware familiarisation.** The available PMA actuators, ESCON 50/5 controller, encoder wiring, STM32 board, bench supply, and mechanical stock were inspected. An existing ESCON configuration was opened in ESCON Studio and preserved as a reference. The result was a safe commissioning sequence: identify hardware, prepare cables, verify wiring, power up under supervision, tune, then integrate STM32. No powered motor test was performed.
- **22 June 2026 – cable preparation.** J1 power leads and J2 motor leads were prepared with suitable terminations, insulation, and planned continuity checks. The system remained unpowered.
- **1 July 2026 – STM32 and ESCON setup.** The STM32 toolchain was verified with a NUCLEO-F446RE blink/button project. A fresh ESCON 50/5 was configured for the PMA-8A actuator with conservative current and speed limits, encoder feedback, closed-loop speed control, Analog Input 1 setpoint, and Digital Input 2 enable. The expected VCC undervoltage fault remained because external motor power was not yet connected.
- **2 July 2026 – supervised power-up.** External power was applied under supervision, the undervoltage fault cleared, automatic tuning completed, and Potentiometer 1 was temporarily used as a low-speed setpoint. First controlled actuator motion was achieved near 100 rpm.
- **3 July 2026 – STM32 integration planning.** The STM32 interface was planned as PA4/A2 for DAC speed setpoint, PC0/A5 for digital enable, PC13 for the user button, and PA5 for LED feedback. The initial generated project workflow caused toolchain confusion, so no STM32-to-ESCON motor test was completed that day.
- **4–5 July 2026 – scope recovery.** The plan was reduced to a minimum viable actuator chain: preserve the potentiometer demo, verify STM32 outputs with a multimeter, connect to ESCON only after verification, and keep a fallback demonstration.
- **6 July 2026 – embedded MVP achieved.** A working CubeIDE project named `new-stm-code` was created. PA4/A2 and PC0/A5 were verified independently with a

multimeter, then wired to ESCON Analog Input 1 and Digital Input 2. Pressing the Nucleo button produced about 1.00 V analog command and 3.31 V enable; ESCON demand and actual speed rose near 100 rpm; the PMA actuator rotated repeatably.

- **7–8 July 2026 – Myo and serial diagnostics.** Laptop serial control, Myo armband acquisition, calibration, template matching, gesture rejection, gyro-based motion rejection, live plotting, CSV logging, and virtual SEA replay were added. Myo diagnostic control worked with the motor disabled.
- **9 July 2026 – finalization.** The repository, report, evidence notes, replay plots, and software checks were finalized. The deliverable was frozen as a physical actuator/embedded-control MVP plus diagnostic Myo subsystem and simulated SEA replay.

2.1 Failure and recovery points

Several useful engineering corrections occurred during the process:

- The first STM32 project attempt did not produce a clean CubeIDE workflow. The recovery step was to regenerate the project as `new-stm-code`, verify that the expected CubeIDE project files existed, and rebuild before touching the ESCON wiring.
- The initial button-control logic behaved opposite to the intended hold-to-run behaviour. This mattered because a wrong default state could command the motor while the button was not pressed. The firmware was therefore tested as a timed blink/interval output first, which separated software logic from human button timing.
- The STM32 outputs were checked with a multimeter before they were connected to the ESCON. This confirmed the three important states: inactive analog output near 0 V, active analog command near 1.00 V, and digital enable near 3.31 V.
- Only after the voltage-only test passed was the STM32 connected to Analog Input 1 and Digital Input 2 on the ESCON. The corrected button firmware then produced the intended hold-to-run behaviour.
- Myo control was kept motor-disabled because the physical SEA output stage did not yet include travel limits, spring deflection sensing, or a finger mechanism. This was a safety decision, not a software limitation.

2.2 Evidence handling

The evidence set contains wiring photographs, multimeter photographs, ESCON Studio screenshots, and short videos of potentiometer-controlled and STM32-button-controlled actuator motion. Large raw phone media were intentionally kept outside Git. The repository includes sanitized report figures, source code, logs, calibration presets, the STM32 firmware tree, and this PDF report. This keeps the repository portable while preserving the evidence chain separately.

Chapter 3

Physical System

3.1 Hardware architecture

The physical test bench comprised a PMA-8A-50-01-E500ML actuator, ESCON 50/5, incremental encoder, NUCLEO-F446RE, and an external DC supply. The ESCON handled encoder-based closed-loop velocity control. The STM32 supplied only the speed setpoint and enable signal.

Table 3.1: Verified STM32-to-ESCON interface.

STM32 signal	ESCON connection	Function
PA4 / A2	J6 pin 1, Analog Input 1+	Analog speed setpoint
GND	J6 pin 2, Analog Input 1–	Analog reference
PC0 / A5	J5 pin 2, Digital Input 2	High-active enable
GND	J5 pin 5, Signal GND	Digital reference

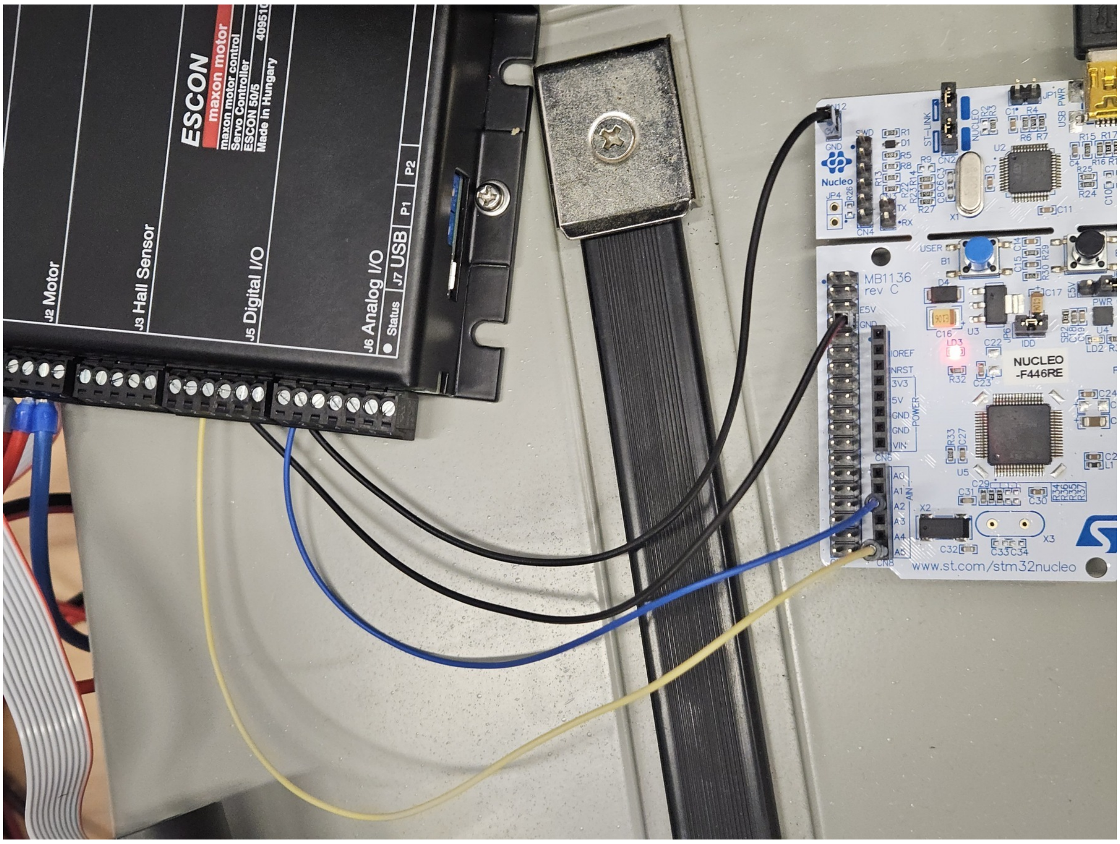

Figure 3.1: Observed STM32-to-ESCON signal wiring. The photograph documents the physical command interface, not a completed finger mechanism.

3.2 ESCON configuration and commissioning

The ESCON was configured as a closed-loop speed controller with a 500-pulse digital incremental encoder. Initial limits were deliberately conservative.

Table 3.2: Commissioning configuration recovered from the final ESCON file.

Parameter	Value
Controller	Maxon ESCON 50/5
Control mode	Closed-loop speed control
Setpoint source	Analog Input 1
Setpoint scaling	0 V → 0 rpm; 1 V → 100 rpm
Enable	Digital Input 2, high-active
Encoder	Digital incremental, 500 pulses/revolution
Nominal current	0.60 A
Maximum output current	1.00 A
Acceleration/deceleration	100 rpm/s
Maximum permissible speed	6000 rpm

Supervised automatic tuning completed successfully. Stable rotation was first observed using Potentiometer 1. The STM32 interface was then verified independently with a multimeter before connection to the ESCON. The measured active outputs were approximately 1.00 V on PA4 and

3.31 V on PC0; inactive output was approximately 0.00 V.

Figure 3.2: Measured STM32 command voltages before connection to the motor controller: inactive DAC (left), active DAC (centre), and active digital enable (right).

Pressing the Nucleo user button asserted enable and the 1 V command; ESCON Studio showed demand and actual speeds near 100 rpm, and the actuator rotated stably. Releasing the button returned the DAC to 0 V, removed enable, and stopped the actuator. Repeated start/stop operation was recorded on video. This is the completed physical MVP.

3.3 Commissioning procedure

The physical commissioning procedure was deliberately ordered from lowest-risk to highest-risk:

1. **Controller-only setup.** The ESCON was connected to the PC over USB and configured while the external DC motor supply was still absent. The visible undervoltage fault was expected at this stage and confirmed that the controller was detected but not yet powered for motion.
2. **Wiring inspection.** Power, motor, and encoder wiring were checked before energizing the system. The PMA actuator remained unloaded and the current and ramp settings were kept conservative.
3. **Automatic tuning.** The ESCON automatic tuning routine was run under supervision. The actuator made only small identification movements, then the controller accepted the tuned speed-loop parameters.
4. **Potentiometer motion test.** The command source was temporarily changed to Potentiometer 1. Digital Input 2 was enabled through a removable jumper, and the potentiometer range was limited to approximately 0–100 rpm. This proved the ESCON, encoder, power wiring, and actuator could produce stable low-speed motion without involving STM32 code.
5. **STM32 output isolation.** The Nucleo was tested separately from the ESCON. A timed output/blink test made PA4 and PC0 alternate between inactive and active states so the multimeter readings could be captured without needing to hold the user button.

6. **Hold-to-run firmware.** After the voltage levels were correct, the firmware was changed back to the intended user-button behaviour: released means STOP, pressed means RUN.

7. **STM32-to-ESCON test.** PA4/A2 was connected to Analog Input 1+, PC0/A5 to Digital Input 2, and the two grounds to the ESCON analog/digital references. Pressing the Nucleo button then produced stable actuator rotation and releasing it stopped the actuator.

Table 3.3: Hardware validation tests and what each test proved.

Test	Method	Conclusion
ESCON detection	USB connection to ESCON Studio	Controller communication was working
Expected undervoltage	ESCON configured before external supply	Fault was understood, not unexplained
Auto-tuning	Supervised ESCON tuning routine	Encoder and motor loop were usable
Potentiometer rotation	Potentiometer 1 setpoint, DI2 jumper enable	Actuator/controller chain moved stably
STM32 inactive output	Multimeter on PA4 and PC0	STOP state produced approximately 0 V
STM32 active output	Timed output and button tests	RUN state produced about 1.00 V and 3.31 V
Button motor test	STM32 wired to ESCON AI1 and DI2	Embedded control MVP was achieved

3.4 Why the blink/interval test mattered

The timed output test was a practical debugging step. During early button testing, the button logic did not behave as intended: the active and inactive states were effectively reversed. Rather than connecting an uncertain signal to the motor controller, the STM32 was programmed to alternate the output state automatically. This made it possible to watch the LED and measure PA4/PC0 with the multimeter at known intervals. Once the inactive and active voltage levels were proven, the code was returned to hold-to-run button control and tested again.

This step is important because it separated three possible faults:

- a firmware logic error in the button state;
- an incorrect STM32 pin assignment;
- an unsafe or unexpected voltage level at the ESCON input.

Only the first issue remained, and it was corrected before connecting the STM32 to the ESCON.

Chapter 4

sEMG Intent Detection

4.1 Acquisition and features

The deprecated Myo armband was connected to a laptop over BLE. The laptop subscribed to eight EMG channels and IMU data. No Bluetooth discovery was required during normal use because the known device address was supplied directly. For each 200 ms sliding window with a 50 ms step, the mean absolute value (MAV) was calculated:

$$f_i = \frac{1}{N} \sum_{n=1}^{N} |e_i[n]|, \qquad i = 1, \ldots, 8. \tag{4.1}$$

Total activity and the normalized spatial pattern were

$$A = \sum_i f_i, \qquad p_i = \frac{f_i}{A + \epsilon}. \tag{4.2}$$

4.2 Calibration and classification

Calibration recorded rest in three forearm orientations, several candidate intents, static rejection gestures, and wrist/forearm motion. Candidate intents were ranked by within-class cosine similarity minus their strongest rejection similarity. The selected pinch attempt was retained only when it produced sufficient activity above rest and a repeatable spatial pattern.

Live classification required activity above the orientation-specific rest threshold, adequate target similarity, a positive margin over the strongest rejection class, and low gyroscope magnitude. Hysteresis and hold times suppressed rapid switching. The detector states were REST, TARGET, GESTURE_REJECT, and MOTION_REJECT. TARGET generated serial command `1`; all other states generated `0`. A three-second STM32 command timeout and repeated STOP commands provided communication fail-safe behaviour.

Calibration presets can be reused, numerically adjusted, or partially retaken. Every run logs timestamped raw EMG, MAV features, similarity scores, state, command, IMU magnitude, and virtual-model outputs to CSV.

4.3 Software implementation

The final software is split into two simple layers.

4.3.1 STM32 firmware

The STM32 firmware implements a small serial command interpreter on the ST-LINK virtual COM port. Its job is not to classify EMG or control position; it only translates a safe high-level command into the two ESCON inputs:

Table 4.1: STM32 serial command behaviour.

Command	STM32 output	Reply
`0` or `STOP`	PA4 = 0 V, PC0 = low, LED off	`ACK STOP`
`1` or `RUN`	PA4 $\approx$ 1 V, PC0 = high, LED on	`ACK RUN DAC=1241 ENABLE=1`
`P 0..100`	Scaled fraction of the 1 V run command	`ACK P=...`
`S` or `STATUS`	No output change	Reports DAC and enable state
Invalid input	Forces STOP	Error message ending in STOP

Boot, parse errors, UART errors, and commands longer than the line buffer all force STOP. A three-second command timeout also forces STOP if the output is active and no fresh command arrives. This keeps the microcontroller behaviour intentionally boring: the laptop may decide when to run, but the STM32 defaults to no analog command and no enable.

4.3.2 Laptop Myo bridge

The Python program performs the higher-level work:

1. connect to the known Myo BLE address;
2. acquire eight-channel EMG and IMU/gyroscope samples;
3. collect or load a calibration preset;
4. compute 200 ms MAV windows with 50 ms step;
5. normalize the eight-channel spatial pattern;
6. compare the live pattern with the target and rejection templates;
7. reject high-gyro motion;
8. print live debug output and write CSV logs;
9. send `1` only for accepted TARGET and `0` for REST or rejection.

The implementation deliberately kept the serial interface binary at the behaviour level: TARGET means RUN, every other state means STOP. More proportional control can be added later, but the present actuator does not yet have a completed finger mechanism or output-position feedback, so a richer command would not be physically meaningful yet.

4.4 Diagnostic Myo validation

The final diagnostic session on 8 July 2026 ran for about 167 s with the motor command disabled. The log contained rest periods, repeated pinch attempts, open-hand rejection, full-fist rejection, wrist flexion/extension, and forearm rotation. Of 1611 logged windows, 384 were classified as REST, 91 as TARGET, 567 as GESTURE_REJECT, and 569 as MOTION_REJECT. This is not presented as a robust clinical classifier; it is evidence that the software can accept the calibrated pinch intent while suppressing many competing gestures and high-motion intervals.

The strongest limitation observed in the log was that pinch attempts can be interrupted by wrist/forearm motion rejection, and open-hand or full-fist templates can sometimes remain close to the target template. Therefore, the correct interpretation is "diagnostic intent interface proven," not "finished proportional hand controller." Motor-enabled Myo testing was deliberately postponed until the mechanism and safety limits are better defined.

Chapter 5

Virtual SEA Replay

5.1 Purpose and assumptions

The virtual model demonstrates the missing compliant mechanism without claiming physical validation. It replays a real recorded Myo session. Pinch intent commands a target of $0°$ and non-target/relaxed state commands $180°$. Motion is bounded to this interval and rate-limited to $90°/\text{s}$.

The virtual motor angle follows the command subject to the rate limit. Finger angle follows motor angle with a first-order time constant $\tau = 0.18$ s. With an assumed 8 mm tendon moment arm,

$$x_m = r(\theta_{max} - \theta_m), \tag{5.1}$$
$$x_f = r(\theta_{max} - \theta_f), \tag{5.2}$$
$$\Delta x_s = |x_m - x_f|, \tag{5.3}$$
$$F_s = \min(k_s \Delta x_s, F_{max}), \tag{5.4}$$

where angular quantities are converted to radians, $k_s = 2.0\,\text{N/mm}$, and $F_{max} = 5\,\text{N}$. These values are adjustable simulation parameters, not identified properties of an assembled mechanism.

5.2 Replay result

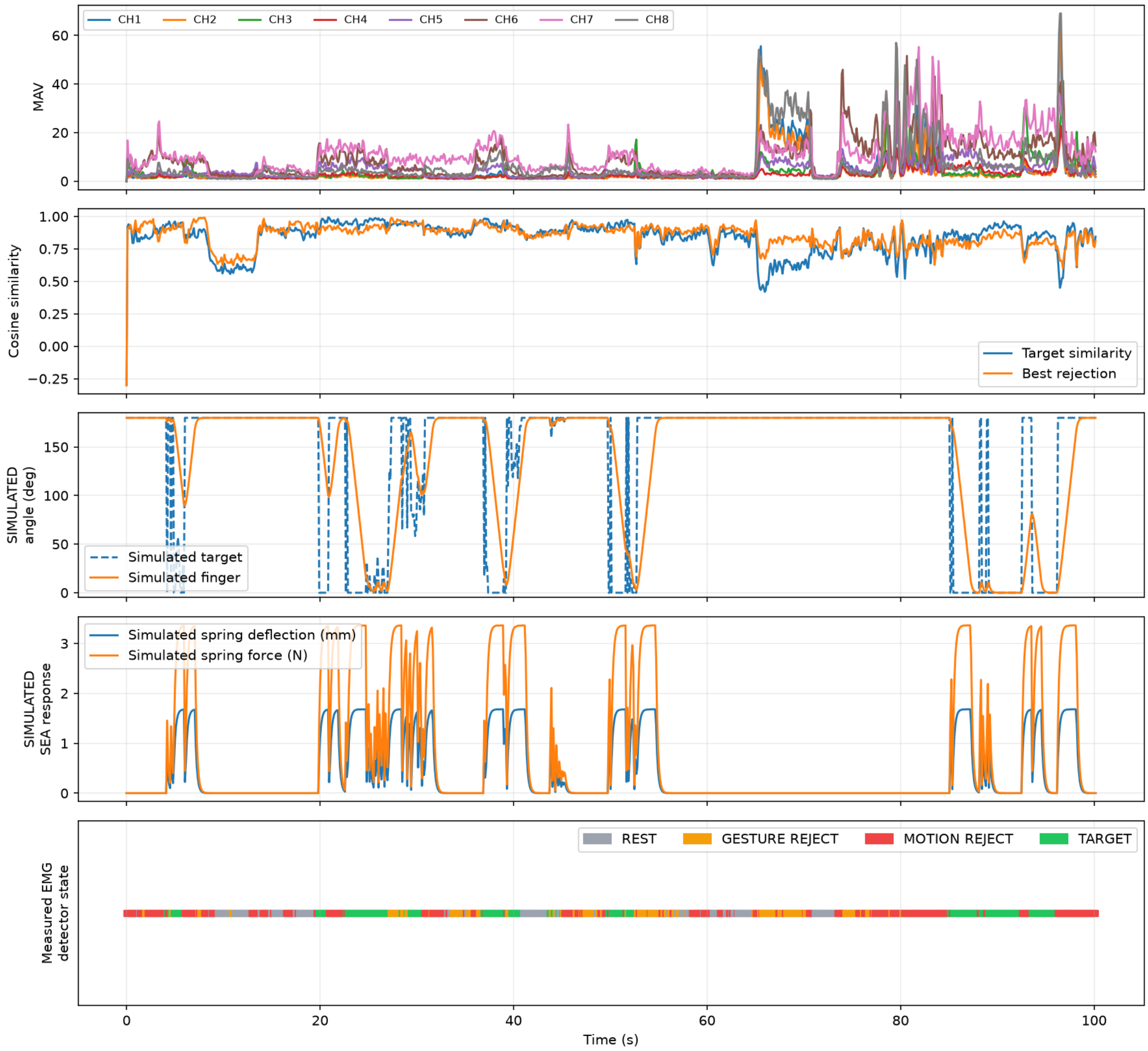

Figure 5.1: Replay of recorded real Myo data. MAV, cosine similarities, and detector states are measured/processed data. Finger angle, spring deflection, and force are explicitly simulated.

The replay shows that accepted pinch intervals produce bounded virtual flexion while rejected gestures and high-motion intervals do not sustain the flexion command. Transient motor–finger displacement produces virtual spring deflection and force. This verifies software data flow and intended qualitative behaviour; it does not validate physical impedance, force accuracy, or human interaction safety.

Chapter 6

Results and Limitations

Table 6.1: Status against the declared study.

Item	Status	Evidence
Actuator and ESCON commissioning	Completed	Tuning, potentiometer motion, monitor screenshots, video
STM32 analog command and enable	Completed	Multimeter measurements, wiring photographs
STM32-controlled motor motion	Completed	Repeated button start/stop video
Myo acquisition and intent detection	Completed diagnostically	Calibration, terminal output, CSV logs, plots
Laptop–STM32 serial protocol	Completed on bench	ACK/STATUS transcript and fail-safe firmware
Physical Myo-to-motor experiment	Not performed	Kept disabled pending controlled safety test
Spring, tendon, and finger mechanism	Not implemented	Components unavailable
Spring-deflection force sensing	Simulated only	Virtual replay
1 kHz physical impedance control	Not implemented	Requires complete SEA sensing and mechanics
EMD/reflex experiment	Not performed	Requires measured spring-force onset and approved protocol

The ESCON is a velocity controller, not an absolute-position controller. Its internal encoder closes the velocity loop but does not provide the STM32 with verified finger angle. Therefore the present physical system cannot safely claim 0–180° finger positioning. A physical SEA requires the compliant element, a calibrated deflection sensor, mechanism geometry, travel limits, and preferably independent output-position sensing.

Chapter 7

Conclusion and Future Work

The project established a reproducible actuator-control baseline and extended it with calibrated Myo intent detection. The physical MVP proves that the STM32 can safely command the ESCON and PMA actuator. The software MVP proves that a repeatable pinch intent can be separated from rest, competing gestures, and high-motion activity for this user. The replay connects recorded EMG intent to an explicitly virtual compliant finger model while preserving the boundary between evidence and simulation.

The next engineering step is mechanical rather than algorithmic: fabricate the 1-DOF finger and tendon path, select and calibrate the spring-deflection sensor, add output travel sensing or hard limits, and only then implement and tune the physical impedance loop. EMD measurement should follow validation of synchronized EMG and measured spring-force onset.

Appendix A

Evidence Matrix

Raw phone media were kept outside the Git repository because the videos are large and may contain private laboratory context. The following table records what the evidence set is meant to support. The video descriptions should be checked against the original clips during final submission packaging.

Table A.1: Evidence items and interpretation.

Evidence	Supports	Interpretation boundary
ESCON Studio screenshots and exported `.edc` file	ESCON 50/5 configuration, encoder mode, Analog Input 1 speed setpoint, Digital Input 2 enable, current/ramp limits	Proves controller configuration, not external mechanism performance
Potentiometer-controlled actuator video	ESCON, encoder, supply, and PMA actuator could produce stable low-speed rotation before STM32 integration	Observed actuator motion only; no finger mechanism attached
Timed STM32 output videos/photos	PA4 and PC0 could be driven in known active/inactive intervals	Debug evidence for voltage-state isolation
Multimeter photographs	PA4 inactive near 0 V, active near 1.00 V, PC0 active near 3.31 V	Electrical command validation before ESCON connection
STM32-to-ESCON wiring photograph	Physical mapping between Nucleo pins and ESCON AI1/DI2 terminals	Documents the command interface wiring
Button-controlled actuator video	Pressing the Nucleo button commanded ESCON motion; releasing stopped it	Completed physical embedded-control MVP
Serial transcript	`STOP`, `RUN`, and `STATUS` commands returned ACKs and changed STM32 state	Bench validation of laptop-to-STM32 protocol
Myo calibration presets and CSV logs	EMG/IMU acquisition, gesture classification, rejection states, and replay data	Diagnostic Myo subsystem only; motor disabled during these runs
Replay figure	Recorded detector state driving a bounded virtual SEA model	Simulated spring/finger response, not measured force or angle

A.1 Raw media inventory

The local raw media transferred from the phone were:

```
videos:
  20260706_101404.mp4  20260706_101655.mp4
  20260706_140056.mp4  20260706_143019.mp4
  20260706_143025.mp4  20260706_143045.mp4

photographs:
  20260706_113957.jpg  20260706_114028.jpg  20260706_114030.jpg
  20260706_121857.jpg  20260706_121859.jpg  20260706_121931.jpg
  20260706_121933.jpg  20260706_142438.jpg  20260706_142951.jpg

configuration:
  STM32_BUTTON_ANALOG_TEST_PMA8A_2026-07-06.edc
```

The report includes only selected normalized figures. The raw media should be submitted separately if the mentor requests full video proof.

Appendix B

Reproduction Commands

Run the Myo system without a motor command:

```
.venv/bin/python run_myo_sea_demo.py \
  --myo-address D4:71:C5:FF:86:F3 \
  --motor disable --mode diagnostic --live-plot
```

Generate the measured/simulated replay figure:

```
MPLBACKEND=Agg .venv/bin/python plot_log.py logs/SESSION.csv \
  --output report/figures/myo-sea-replay.png
```

Run the software checks:

```
.venv/bin/python -m unittest -v
.venv/bin/python sea_model.py
```

Bibliography

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[3] Maxon Motor AG, *ESCON 50/5 Hardware Reference*, document 409510.